

IMPLEMENTATION OF DATA HIDING METHODS FOR RESOURCE CONSTRAINED IoT DEVICES

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Abstract

The rapid proliferation of Internet of Things (IoT) devices across industrial, domestic, and commercial environments has brought data security to the forefront of embedded systems research. Conventional encryption algorithms such as AES and RSA, while cryptographically robust, impose computational and memory overheads that are incompatible with resource constrained microcontrollers. This project addresses this gap by designing, implementing, and evaluating a tri-modal lightweight encryption framework on an ESP32 microcontroller paired with an SSD1306 OLED display, programmed entirely using the Arduino IDE. The first cipher algorithm is Position-Variant Radial Stick Cipher (PVRSC). This algorithm uses a method that replaces the plaintext characters with geometric symbols called radial sticks, which can be drawn directly onto an OLED screen. The second cipher algorithm is the Magic Square Zigzag Cipher (MSZC). In this algorithm, the characters in the plain text are converted to ASCII numbers and placed within an odd order magic square that is automatically created for each individual ciphertext. The last proposed algorithm is the Fractal Substitution Cipher (FSC). In this algorithm, a multi-level substitution approach based on fractals is used. For each of these approaches, the printable ASCII range is repeatedly divided into subranges that will then be reversed to create the substitution mapping tables. All the above modules are programmed using C++ language on the ESP32 device via the Arduino development environment. Entropy, avalanche effect, NPCR, UACI, and correlation tests have been successfully passed for all three cipher methods, showing their cryptographic performance close to the theoretical limits. care.